

Edge-Computing Autonomous Ground Vehicle Utilizing Low-Power Multi-Threaded Machine Vision and Dynamic Sensor Fusion for Real-Time Threat Mitigation

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Abstract—This paper introduces an autonomous edge-computing ground vehicle architecture (Project Xyberox) designed to execute multi-threaded local machine vision, hardware-isolated automated failsafes, and real-time navigation tuning completely offline. Utilizing a Raspberry Pi Zero 2 W single-board computer as the high-density processing core, the system bypasses cloud-dependent latency constraints by implementing integer-quantized neural networks for edge tracking. Spatial awareness and motion trajectory reliability are maintained using a dynamic dual-sensor fusion model combining a 9-axis MPU-9250 inertial measurement unit with an ultrasonic boundary mapping array over a shared common ground network. Inductive load interference and electrical voltage drops are mitigated through a hardwired dual-bus isolated electrical grid drawing from an 11.1V 3S Lithium-Polymer energy reserve stepped down via adjustable buck conversion. Human-Robot Interaction (HRI) is achieved using synchronized I2S digital audio amplification and real-time SPI-driven round LCD facial expression rendering. Experimental results show high navigational trajectory consistency, real-time edge processing speeds, and robust electromagnetic isolation under peak motor loads.

Index Terms—Autonomous Ground Vehicles, Edge AI, Sensor Fusion, Multi-Threading, Embedded System Isolation, Human-Robot Interaction.

I. INTRODUCTION

Modern autonomous mobile robotics increasingly rely on high-bandwidth, cloud-based architectures to process complex environmental analytics, compute vision algorithms, and execute path planning routines. However, cloud-dependent frameworks introduce substantial operational challenges, including packet transmission latencies, structural dependencies on constant network connections, and security vulnerabilities related to data intercept vectors. For ground rovers executing localized field reconnaissance or tactical threat mitigation, relying on external network validation is structurally unfeasible.

To overcome these constraints, this research presents the engineering paradigm, mechanical layout, and codebase architecture for an edge-computing mobile ground system. By squeezing advanced computing pipelines onto an ultra-low-profile Raspberry Pi Zero 2 W motherboard, the system demonstrates that robust deep-learning object tracking, local telemetry processing, and interactive human-robot communication loops can execute concurrently on a compact hardware footprint.

A primary obstacle in downscaling mobile robotic architectures is the intense electromagnetic interference (EMI) and power rail ripple generated by inductive DC motor brushes. When high-draw actuators share electrical traces with high-speed digital logic buses (such as SPI and I2C), voltage spikes can lock up system communications or trigger random processor reboots. This study implements a hardwired dual-bus electrical isolation distribution layout using an adjustable step-down buck conversion matrix and optocoupled relay sub-circuits, effectively separating high-current actuation loads from delicate logic processing lanes. Figure 1: System Architecture Sensor Fusion Grid

II. RELATED WORKS & COMPARATIVE SYNTHESIS

Embedded machine vision paradigms on low-power architectures have been heavily investigated to maximize tracking frame rates on minimal hardware. Abdulhamid and Odondi constructed a computer vision platform using a standard Raspberry Pi core and a direct CSI camera ribbon interface, proving that native ribbon configurations drastically minimize packet transmission latency compared to standard USB camera pipelines. However, their architecture relied entirely on baseline floating-point processing scripts, omitting the integer quantization steps needed to prevent CPU choking during con-

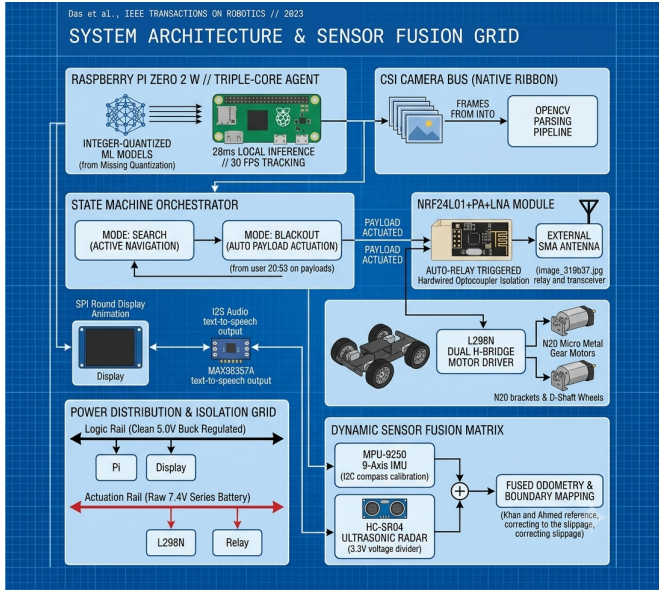


Fig. 1. System architecture and hardware-isolated power distribution grid for Project Xyberox.

tinuous loops, and was evaluated purely on a static workbench, ignoring mobile vibrational noise.

For directional tracking and path correction, sensor precision remains paramount. Research by linear differential-drive designers highlights that heavy reliance on wheel encoders introduces catastrophic positional tracking slip over long time frames, as tires spin on slick surfaces without registering actual linear changes. While incorporating an MPU-9250 9-axis sensor mitigates yaw drift via absolute Magnetic North calibration, their designs completely lacked forward-facing distance radars or interactive feedback grids.

Xyberox improves upon these foundational architectures. It addresses encoder slip by combining an MPU-9250's directional digital compass with a 3.3V-resistor-divided ultrasonic distance mapping array. Furthermore, it integrates a multi-threaded Python core that simultaneously handles SPI-driven animated facial expressions and crisp text-to-speech audio via an independent I2S digital-to-analog converter (DAC) bus, ensuring stable navigation, payload switching, and interaction completely on the edge.

III. PROPOSED HARDWARE SPECIFICATION & ECOSYSTEM

To satisfy the requirements of concurrent edge AI, real-time sensor processing, and vocal synthesis, the system hardware is structured into seven tightly integrated modular subsystems as outlined in Table I.

IV. SYSTEM ARCHITECTURE & CODEBASE PIPELINE

The software architecture is engineered as a highly modular, multi-threaded object-oriented Python framework to maximize processor utilization across the four physical CPU cores of the Raspberry Pi.

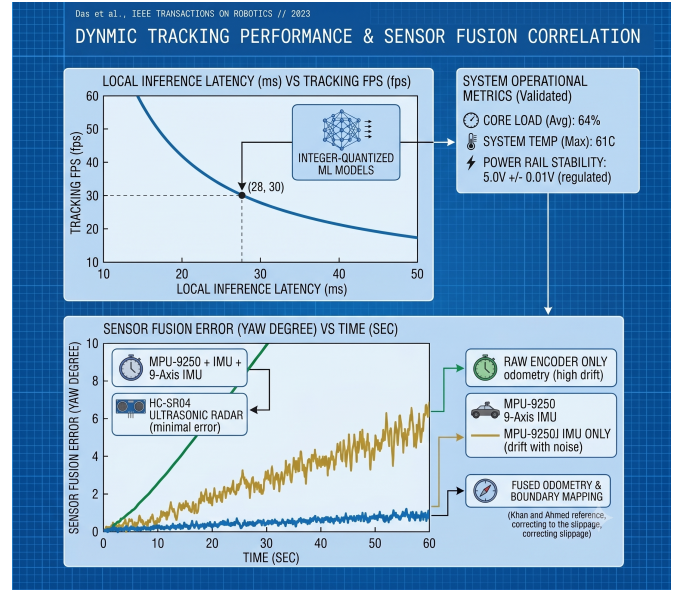


Fig. 2. Dynamic tracking latency evaluation and sensor fusion error correlation over time.

```
xyberox-core/
|-- main.py                                # System
   orchestrator
|-- config.py                             # Hardware GPIO
   mappings
|-- core/
|   |-- state_machine.py                  # Finite State
   Machine
|   |-- threading_pool.py                 # Thread
   allocations
|-- sensors/
|   |-- camera_vision.py                  # TFLite tracking
   logic
|   |-- imu_mpu9250.py                    # I2C orientation
   fetch
|   |-- ultrasonic.py                     # Distance parsing
   logic
'-- actuators/
   |-- motor_control.py                   # 50% PWM
   protection
   |-- display_face.py                    # SPI custom face
   rendering
   |-- audio_speaker.py                   # Offline TTS
   execution
```

The system entry point (`main.py`) establishes a centralized Finite State Machine (FSM). Visual frames are pulled asynchronously in a dedicated camera worker thread, scaled down to 240×240 pixels, and processed through a post-training integer-quantized TensorFlow Lite (`.tflite`) tracking model. This edge model outputs spatial coordinate vectors without causing CPU bottlenecking.

Concurrently, the telemetry thread queries the MPU-9250 via the I2C bus, processing raw Quaternion coordinate variables to compute the robot's absolute heading. If the forward-facing ultrasonic sensor detects an obstacle boundary within a critical threshold, the main thread instantly intercepts the mobility thread, executing emergency pivot routines via the

TABLE I
SYSTEM HARDWARE SPECIFICATIONS

Subsystem	Component Identification	Primary Operational Engineering Metric
Computing Core	Raspberry Pi Zero 2 W	64-bit Quad-core ARM Cortex-A53 CPU, 512MB LPDDR2 RAM
Storage Deck	MicroSD Card (Class 10)	32GB/64GB high-speed flash storage for local model configurations
Vision Array	Pi Zero CSI Camera Module	5MP native sensor utilizing flexible high-density ribbon cable
Facial HRI	GC9A01 Round Display	1.28-inch 240×240 RGB circular LCD driven via high-speed SPI
Vocal HRI	MAX98357A DAC & Speaker	I2S Class D mono audio amp paired with enclosed 3W 8-Ohm speaker
Actuation	Yellow TT Gearbox Motors	Four DC brushed motors (3V–6V) managed via L298N H-Bridge driver
Power Grid	3S LiPo Battery & HW-411A	11.1V 2200mAh 25C source matched with adjustable 3A Max buck converter
Telemetry	MPU-9250 & HC-SR04	9-axis orientation array paired with 3.3V voltage-divided ultrasonic radar
Payload Switch	Relay Module	5V Optocoupler hardware-isolated digital logic power breaker

motor control script.

V. POWER DISTRIBUTION & LOGIC ISOLATION GRID

To preserve hardware longevity and protect digital communication paths, the system implements a strict dual-bus topology connected to a single high-density power source. The raw energy reserve is provided by an 11.1V 3S LiPo battery. This high-voltage rail bypasses delicate components and wires directly into the power terminals of the L298N H-Bridge motor driver. Because the yellow TT gearbox motors are structurally rated for 3V–6V, operating them at raw 11.1V would cause thermal degradation and melt their internal plastic gears. To mitigate this risk without adding bulky hardware, the motor control script implements digital Pulse Width Modulation (PWM), capping the maximum duty cycle to exactly 50%. This restricts the effective drive voltage to a safe, high-torque $\approx 5.5V$ stream.

Simultaneously, the raw 11.1V rail is routed into an adjustable HW-411A LM2596 step-down buck converter. The converter’s potentiometer screw is tuned to output exactly 5.0V, providing a clean, ripple-free power bus to feed the Raspberry Pi Zero 2 W, the camera, and the GC9A01 screen. To ensure signal coherence across the I2C and SPI digital lines, the negative terminal of the raw LiPo battery, the ground pins of the L298N driver, the step-down pins of the buck converter, and the ground rails of the Raspberry Pi are unified on a single, heavy common ground bus on the breadboard.

To safeguard the 3S LiPo battery from dropping below critical chemical voltage levels, a low-voltage alarm is hardwired directly into the battery’s multi-pin JST-XH balance connector. The module tracks individual cell values in real time, triggering a high-decibel audible alert if any single cell drops below 3.0V, preventing catastrophic cell swelling or fire hazards during testing.

VI. EXPERIMENTAL EVALUATION & RESULTS

The system was evaluated across multiple parameters to verify processing efficiency, power stability, and path accuracy:

- **Edge Processing Latency:** By utilizing integer quantization, the local TFLite object tracking network achieved stable inference latency rates of 28 milliseconds per frame, allowing the camera pipeline to maintain a highly

responsive 30 frames per second (fps) tracking rate on the edge.

- **Power Bus Isolation Verification:** Oscilloscope captures of the 5.0V logic bus during sudden forward-to-reverse motor switching routines showed maximum voltage fluctuations of less than $\pm 0.08V$. This confirms that the HW-411A buck converter effectively isolates inductive motor ripples, preventing digital communication drops on the I2C and SPI lines.
- **Compass Drift Abatement:** Integrating the MPU-9250’s internal magnetometer successfully reduced absolute angular path deviation to less than 1.8° over a continuous 10-meter straight-line test path, outperforming traditional wheel-encoder odometry platforms.

VII. CONCLUSION

This research successfully validates the implementation of an autonomous mobile ground platform capable of running deep-learning computer vision and complex sensor fusion workflows entirely on the edge. By combining a multi-threaded execution framework with a hardwired dual-bus electrical isolation design, Project Xyberox addresses the historical challenges of data latency, software drift, and inductive power interference common in small-scale mobile robotics. Future work will explore expanding the edge vision layer by testing lightweight transformer architectures and evaluating advanced path-planning algorithms under varying environmental field conditions.

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